

Spot the Problem: Naive Comparisons and Reading an RCT

Module 2 · Day 1, Session 3: Opening the Box on Module 1's Numbers

3ie

0 · Where Module 1 left us

Where we left GreenWaste

In Module 1 you met the **GreenWaste Program** and saw three estimates of what it did to businesses' waste costs. The bar for a national roll-out, fixed for the whole week: **cut at least 1,000 AED.**

From memory: which one did Module 1 tell you to trust, and what was wrong with the other two?

Before and after	-665 AED
Enrolled vs not	-1,447 AED
Randomised	-1,014 AED

Module 1 called the first two **counterfeit counterfactuals**: one ignores what else changed over time, the other compares groups that were never alike.

Today we open the box

Module 1 showed you the three numbers. Today we compute them from the raw data, on screen, and look for each problem in the data itself.

- 1 Meet the data: what one row is, and who was offered what
- 2 Rebuild the two counterfeit numbers, then catch each one out
- 3 Run the lottery ourselves, and see why it works
- 4 Read the randomised result against the 1,000 AED rule

You will not write any code. You will call the numbers; I will type them in and press **Run**. The grey boxes are real R, running in this page.

1 • Meet the data

One row, one business, one round

R Code

Start Over

Run Code

```
1 rows <- gw[, c("business_identifier", "round", "treatment_neighborhood",
2               "enrolled", "waste_management_costs")]
3 round(head(rows, 4))
```

business_identifier	round	treatment_neighborhood	enrolled	waste_management_costs
5	0	1	1	1519
5	1	1	1	1958
11	0	1	1	1308
11	1	1	1	240

round 0 before, 1 after.

treatment_neighborhood 1: the neighbourhood was offered the programme.

enrolled 1: the business took part.

The file has 19,826 rows. How many **businesses**?

Half: 9,913 businesses, each measured twice.
One row is a business in a round.

Who was offered, who took part?

R Code

Start Over

Run Code

```
1 base <- subset(gw, round == 0)
2 base$businesses <- 1
3 aggregate(cbind(businesses, enrolled) ~ treatment_neighborhood + eligible,
4           data = base, FUN = sum)
```

treatment_neighborhood	eligible	businesses	enrolled
0	0	2290	0
1	0	1995	0
0	1	2664	0
1	1	2964	2964

Who enrolled in the offered neighbourhoods? And in the others?

In offered neighbourhoods **every eligible business enrolled, and no other did.** So “did not take part” means “scored above the cut-off”: a different kind of business.

2 · Rebuild the two counterfeit numbers

Comparison A: before and after

Module 1 said **-665**. Before I press Run: will the raw rows give the same number?

R Code

↺ Start Over

▶ Run Code

```
1 before <- mean(gw$waste_management_costs[gw$enrolled == 1 & gw$round == 0])
2 after  <- mean(gw$waste_management_costs[gw$enrolled == 1 & gw$round == 1])
3 round(data.frame(before, after, change = after - before))
4 bars(Before = before, After = after)
```



Comparison B: with and without

R Code

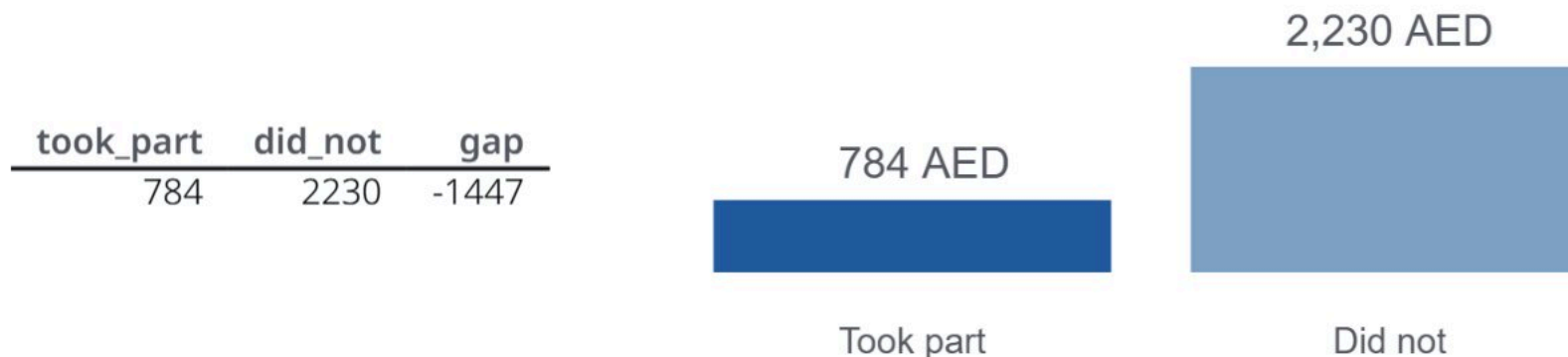
Start Over

Run Code

```

1 offered <- subset(gw, treatment_neighborhood == 1 & round == 1)
2 took    <- mean(offered$waste_management_costs[offered$enrolled == 1])
3 didnt   <- mean(offered$waste_management_costs[offered$enrolled == 0])
4 round(data.frame(took_part = took, did_not = didnt, gap = took - didnt))
5 bars("Took part" = took, "Did not" = didnt, col = c(ACCENT, ACCENT2))

```



Same neighbourhoods, after the programme. Module 1 said **-1,447**. Two readings of one programme, **-665** and **-1,447**: one misses the 1,000 AED bar, one clears it.

[MENTI PLACEHOLDER] Which way is each one wrong?

You know from Module 1 both numbers are biased. The harder question:

Is 665 too big or too small?

Is 1,447 too big or too small?

Go to **menti.com** and enter code **1234 5678**, or scan the code.



3 · Catch each one out, in the data

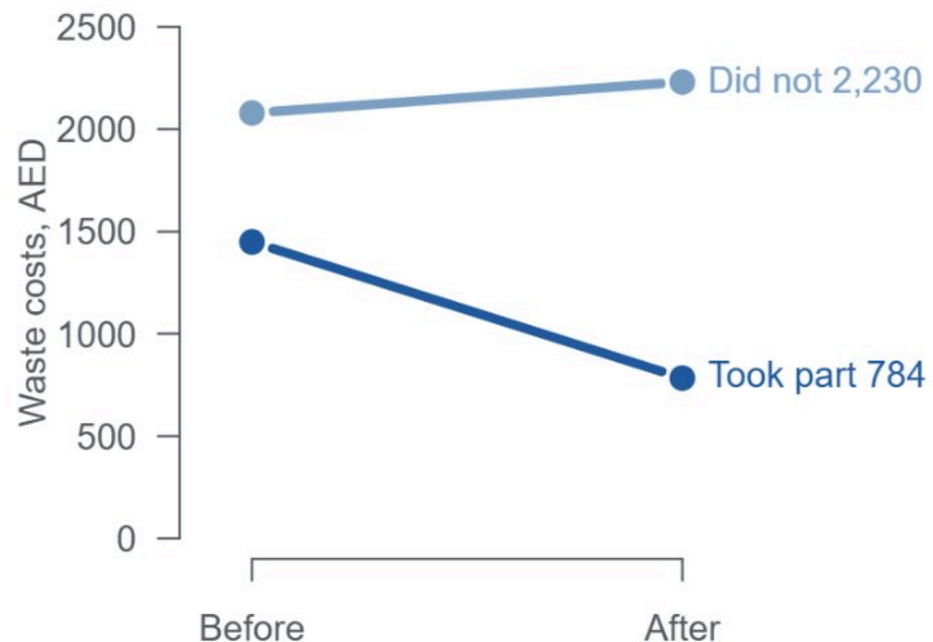
A: what happened to everyone else?

R Code

↺ Start Over

▶ Run Code

```
1 plot_two_lines()
```



Before-and-after compares the programme to **the past**. If costs were moving anyway, that movement is baked into the 665.

Look at the businesses that did **not** take part. Which way did their costs go with no programme at all?

Up, by about 150 AED. Costs were rising anyway, so the 665 fall **understates** what the programme did.

B: the same gap, before the programme existed

- After the programme
- Before the programme

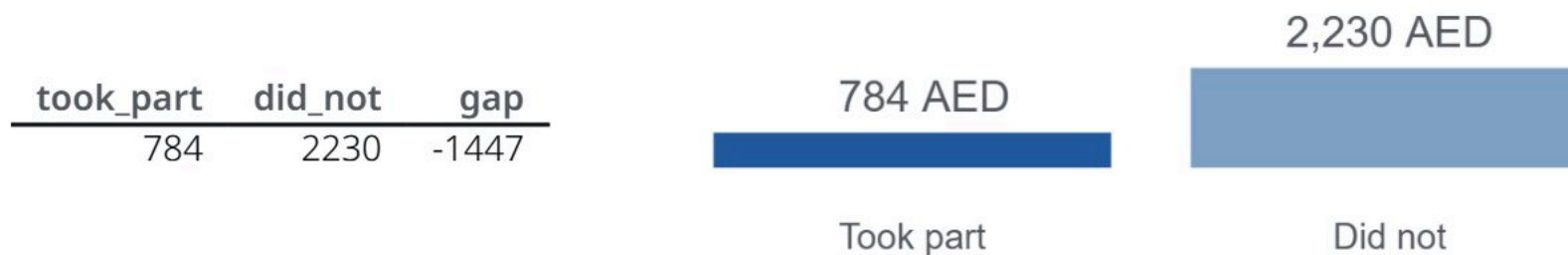
Before the programme, nobody had received anything.
What gap should we see?

R Code

↺ Start Over

▶ Run Code

```
1 r <- if (when == "Before the programme") 0 else 1
2 offered <- subset(gw, treatment_neighborhood == 1 & round == r)
3 took <- mean(offered$waste_management_costs[offered$enrolled == 1])
4 didnt <- mean(offered$waste_management_costs[offered$enrolled == 0])
5 round(data.frame(took_part = took, did_not = didnt, gap = took - didnt))
6 bars("Took part" = took, "Did not" = didnt, col = c(ACCENT, ACCENT2))
```



It should be zero. It is **-630** AED: a head start, so the 1,447 gap **overstates** the effect.

Why they were already different

R Code

↺ Start Over

▶ Run Code

```
1 before <- subset(gw, round == 0 & treatment_neighborhood == 1)
2 round(aggregate(cbind(staff_size, business_area, age_manager,
3                       advanced_filtration) ~ enrolled, data = before, FUN = mean), 2)
```

enrolled	staff_size	business_area	age_manager	advanced_filtration
0	4.30	2.60	52.03	0.46
1	5.77	1.68	41.66	0.72

Same businesses, **before** the programme.
Enrolled (1) vs not (0) on four things we measured.

Younger managers, smaller premises, more filtration already installed. The two groups were different kinds of business, and **some of the ways they differ we did not measure at all.**

Two biases, opposite directions

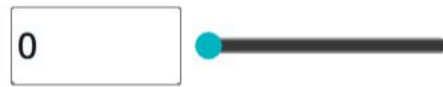
- Before and after** -665 AED · folds in the rising costs everyone faced · UNDERSTATES
- With and without** -1,447 AED · folds in a 630 AED head start · OVERSTATES
- So the truth** sits somewhere between the two, and neither comparison can say where

Check your Menti answer. What would it take to land on the truth **deliberately**, instead of guessing between two wrong numbers?

In both cases **the two groups were not comparable**. After the break we look at a design that makes them comparable by construction.

The GreenWaste lab

Costs rise for everyone anyway (AED)



Joiners are smaller businesses (AED)



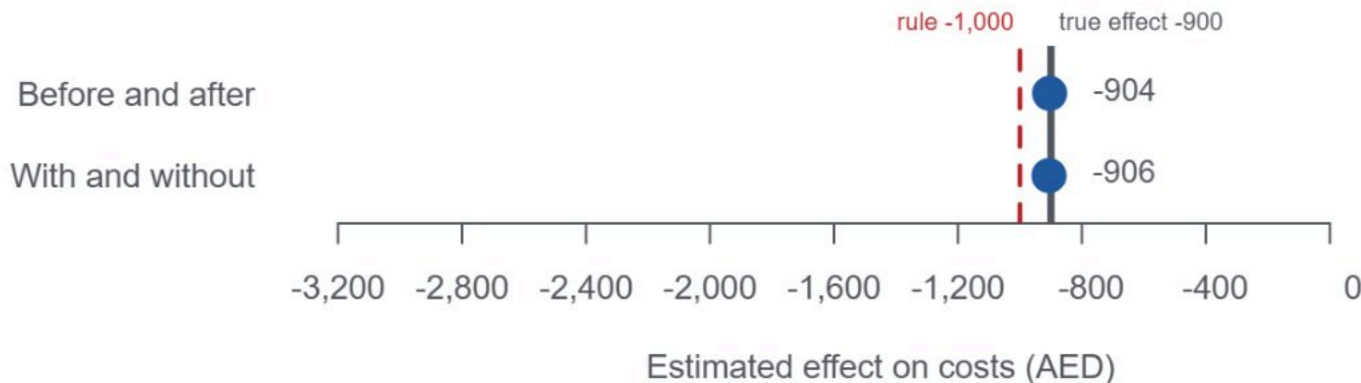
Joiners differ in something nobody measured (AED)



An invented GreenWaste of 20,000 businesses where we know the truth: taking part cuts costs by **900 AED**. Each slider switches on one problem.

Before each move: **which estimate will that slider pull away from the truth, and which way?**

```
lab(c("before_after", "with_without"), lab_trend, lab_size, lab_hidden)
```



Break

Five-minute break

Back in five minutes. When we return, we run the lottery ourselves.

4 • Run the lottery yourself

What the lottery buys you

Module 1 recap. GreenWaste was offered to **neighbourhoods drawn by lot**. Inside both kinds of neighbourhood, only eligible businesses count.

So the randomised comparison is: eligible businesses in drawn neighbourhoods vs eligible businesses in the others.

The claim is that a draw makes the two groups **alike on average**, on what we measured *and* on what we did not.

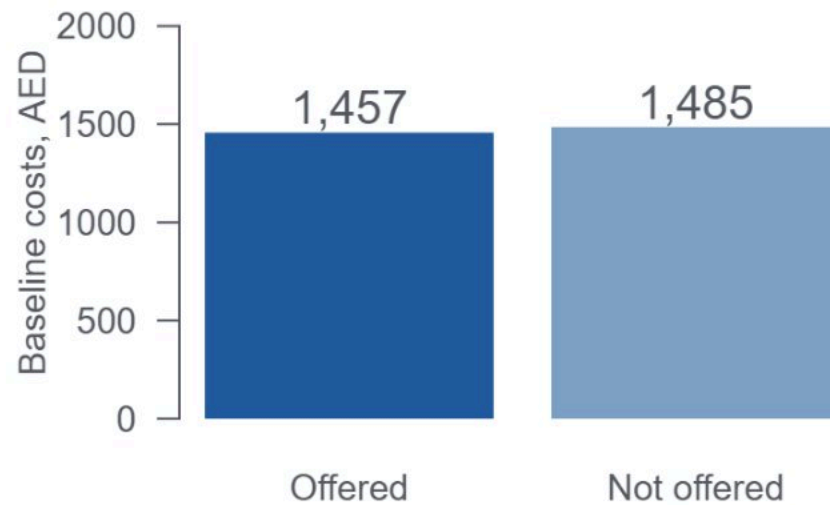
If the claim holds, what should the gap in **baseline** costs between drawn and not-drawn neighbourhoods look like?

Draw the lottery again

How neighbourhoods are chosen ☒ Random draw
☐ Officials pick

Draw again

```
draw_lottery(how, seed = draw)
```



Gap BEFORE the programme started: -28 AED

Bars show **baseline** costs, before anyone was offered anything.

Press **Draw again** five times. How big does the gap get? Then let officials pick.

When officials pick, the offered neighbourhoods start hundreds of AED ahead, so the programme looks successful before it has done anything.

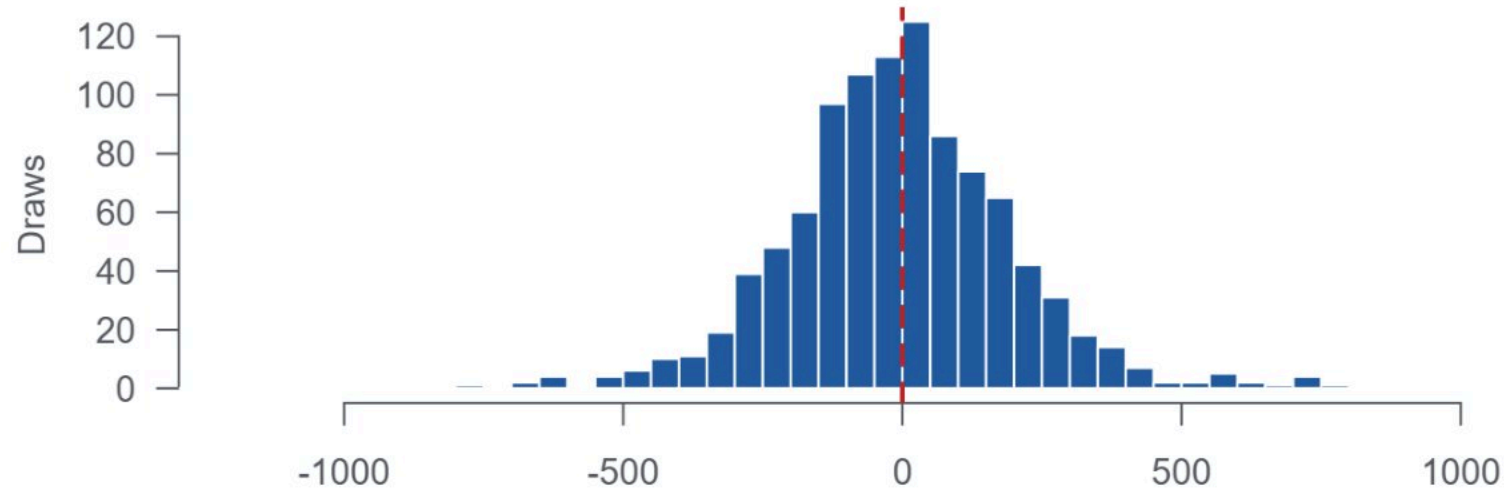
Two neighbourhoods or two hundred?

Neighbourhoods in the lottery

4



`crowd_gaps(k)`



4 neighbourhoods: in 95% of 1000 lotteries the groups start within +/- 404 AED

One thousand lotteries at each size. Start at 4 neighbourhoods, then slide up. **At what size would you trust a single draw?**

[MENTI PLACEHOLDER] Would you trust this draw?

Suppose GreenWaste had drawn only **6 neighbourhoods**, 3 offered and 3 not. Before the programme the offered three already had costs **300 AED lower**. Afterwards they are **900 AED lower**.

What do you conclude? **A** The effect is 900: it was randomised. **B** About 600: take off the head start. **C** Too few neighbourhoods to trust either number. **D** I need to see something else first. Vote, then say what you would need.

C, with D's instinct. A draw makes groups alike **on average over many draws**, not in every draw: with a handful of units a 300 AED head start is ordinary (the slider showed about ± 400 at 4 neighbourhoods). B looks careful but assumes the head start would have stayed the same. Ask how many units were randomised, and for the balance table.

The lab again: add the lottery

Costs rise for everyone anyway
(AED)



Joiners are smaller businesses
(AED)



Joiners differ in something
nobody measured (AED)



Move any slider as far as you like. **Which estimate never moves, and why not?**

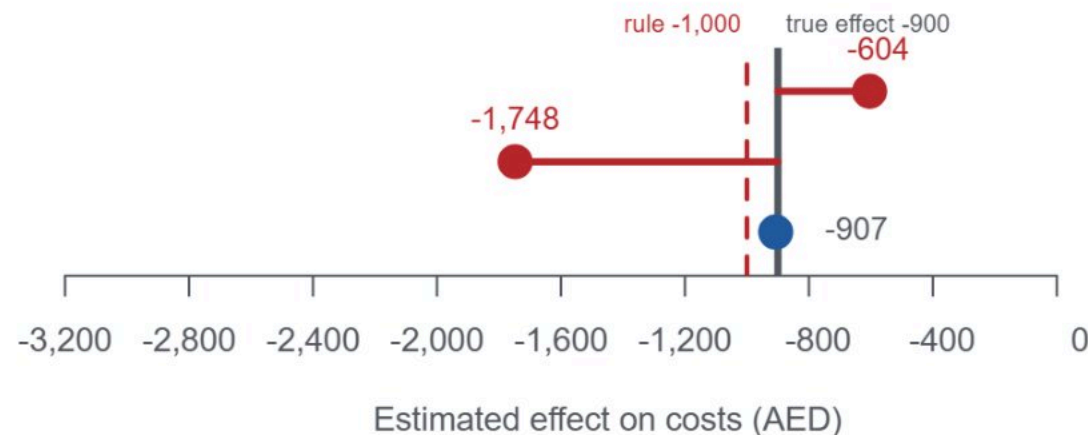
The randomised one. The lottery decides who is offered, so every problem lands equally on both sides of the comparison and cancels out.

```
lab(c("before_after", "with_without", "randomised"), lab2_trend, lab2_size, lab2_hidden)
```

Before and after

With and without

Randomised



5 · Reading the randomised result

First check: did the real draw work?

R Code

↺ Start Over

▶ Run Code

```
1 before <- subset(gw, round == 0 & eligible == 1)
2 round(aggregate(cbind(waste_management_costs, staff_size, business_area,
3                       age_manager) ~ treatment_neighborhood, data = before, FUN = mean), 1)
```

treatment_neighborhood	waste_management_costs	staff_size	business_area	age_manager
0	1457.4	5.7	1.7	42.3
1	1449.0	5.8	1.7	41.7

Same check as three slides ago, but now the two groups are **drawn** neighbourhoods (1) vs the others (0), eligible businesses only.

Compare with the enrolled vs not table: there, managers differed by ten years. Here, the groups look alike **before** the programme. That is what a balance table is for.

The effect is a difference of two averages

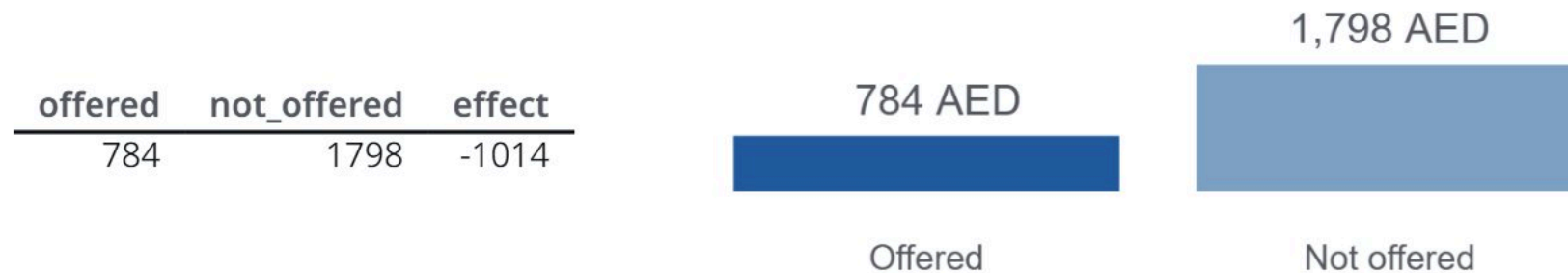
Module 1 said **-1,014**. Three lines of averages. Will they match?

R Code

↻ Start Over

▶ Run Code

```
1 after <- subset(gw, round == 1 & eligible == 1)
2 offered <- mean(after$waste_management_costs[after$treatment_neighborhood == 1])
3 not_offered <- mean(after$waste_management_costs[after$treatment_neighborhood == 0])
4 round(data.frame(offered, not_offered, effect = offered - not_offered))
5 bars(Offered = offered, "Not offered" = not_offered, col = c(ACCENT, ACCENT2))
```



Yes. The draw made the groups alike, so the gap between the two averages **is the effect**, with no adjustment.

The regression gives the same number

R Code

↺ Start Over

▶ Run Code

```
1 fit <- lm(waste_management_costs ~ treatment_neighborhood, data = after)
2 round(data.frame(coefficient = coef(fit)))
```

	coefficient
(Intercept)	1798
treatment_neighborhood	-1014

The table an evaluator hands you comes from a line like this one.

(Intercept) is the not-offered average.

treatment_neighborhood is the gap between the groups.

Which of these two numbers is the impact? And what is the other one?

The coefficient is the difference you just computed by hand. The intercept is where the comparison group sits: **not an effect**. Reading it as one is a common mistake, by people and by AI.

How sure? Count neighbourhoods, not businesses

R Code

↺ Start Over

▶ Run Code

```

1 library(estimatr)
2 fit_nb <- lm_robust(waste_management_costs ~ treatment_neighborhood,
3                   data = after, clusters = neighborhood_identifier)
4 round(as.data.frame(rbind(
5   "as if 5,628 separate draws" = confint(fit)["treatment_neighborhood", ],
6   "counted by neighbourhood"   = confint(fit_nb)["treatment_neighborhood", ])))

```

	2.5 %	97.5 %
as if 5,628 separate draws	-1055	-974
counted by neighbourhood	-1093	-935

The lottery drew about 200 **neighbourhoods**, not 5,628 businesses, and businesses in one neighbourhood are not independent pieces of evidence.

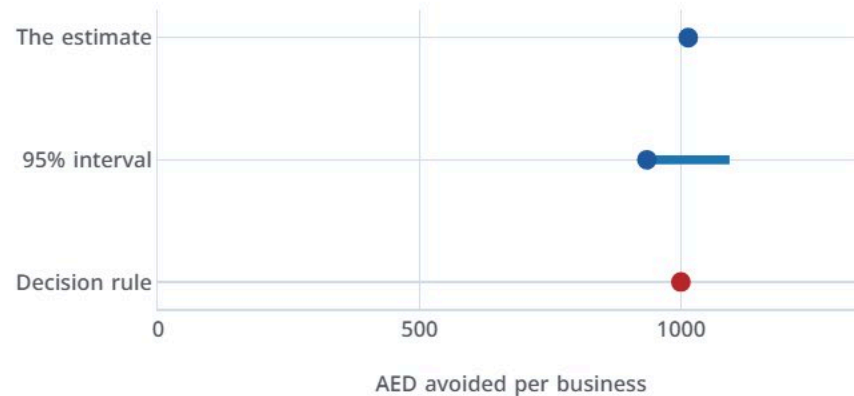
Counting neighbourhoods **widens** the interval: its least generous end moves from 974 to **935** AED. Ask: *"were your errors clustered where you randomised?"*

Which number is the impact?

- The impact** the treatment_neighborhood coefficient: -1,014 AED · the difference the programme made
- Not the impact** the intercept, 1,798 AED · where the comparison group sat
- Not sufficient either** the p-value · it says the effect is unlikely to be zero, not that it is big enough

A summary reads: *“highly significant, so the programme works and should be scaled.”* Which line of this table does it misuse?

Does it clear the rule?



-1,014

AED, the randomised estimate

The rule **1,000** Least it could be **935**

The estimate clears the bar. The interval **spans** it: at its least generous the programme avoids 935 AED, short of 1,000.

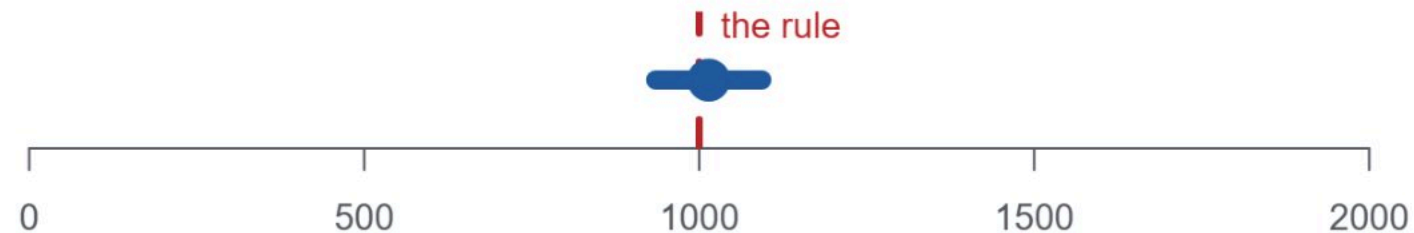
Would you sign a national roll-out on this? Commit, then we ask what a smaller trial would have told us.

What if the trial had been smaller?

Neighbourhoods in the trial

196

`small_trial(kt)`



196 neighbourhoods: avoids 1014 AED, interval 935 to 1093

Slide down to 20. **What happens to the estimate, and to the interval?**

The estimate barely moves, but the interval widens to about 590 to 1,290 AED. At that size the trial cannot tell you whether the bar is cleared.

6 • Put AI to work

AI Snapshot: a paragraph for the minister

Groups, paste this prompt, written the way a busy official might write it:

“Our pilot cut waste costs by 665 AED per business (costs before vs after, $p < 0.001$). Write a short paragraph for the minister on what this shows.”

Mark anything the evidence does not support. Three minutes, in your groups. Then run the same prompt yourselves: did your AI write something similar?

A response like this is possible, especially with a short or leading prompt:

The GreenWaste pilot produced a clear result: waste-management costs fell by 665 AED per business, and **with $p < 0.001$ we can be confident the programme caused this reduction. Because the same businesses were measured before and after, differences between businesses are already accounted for, so the 665 AED saving can be attributed to the programme. This makes a strong case for national scale-up.** As with any pilot, results should be monitored as the programme expands.

Debrief: four errors

1 • Significant is not causal. $p < 0.001$ says the fall is unlikely to be chance. It says nothing about *what* caused it.

2 • Half right. Following the same businesses removes fixed differences between them. It does not remove what changed for everyone: costs for businesses that did not take part rose by about 150 AED.

3 • It reported 665 as the effect. That is the naive figure. The randomised estimate is 1,014 AED, and the prompt gave the model no way to know that.

4 • It recommended scale-up without asking what bar the programme has to clear, or what 665 was compared to. The closing caveat is generic.

The prompt steered the answer. It supplied a p-value, the word “cut” and a request for the minister, so a model may write a confident summary instead of questioning the design. People asked for a quick paragraph for a minister often do the same.

The same question, a better prompt

What people actually type

Our pilot cut waste costs by 665 AED per business (before vs after, $p < 0.001$). Write a short paragraph for the minister.

The prompt frames the fall as the programme's doing, and the model may follow that framing.

What to type instead

Costs for participating businesses were 665 AED lower after the pilot than before ($p < 0.001$). We also have businesses that did not take part and a randomised comparison. The scale-up rule is at least 1,000 AED. Before writing anything, tell me what this before-and-after number can and cannot show. **Ask me questions before you answer.**

Run both. Which one questions the 665 before writing the paragraph? What did each explain well?

Two habits: make the AI **ask you questions first**, and have a **second chat** review the first one's answer.

7 • Close

Your turn: ask AI first

Before we show our list of questions to ask someone presenting an RCT, ask an AI tool:

“I am a commissioner about to hear an evaluator present a randomised controlled trial (RCT) result. What are the key questions I should ask?”

Three minutes. Then compare with our list on the next slide: what did it miss, what did it add, and is anything incorrect?

Three questions to ask the evaluator

- 1 Compared to what? Before-and-after, with-and-without, or a fair comparison?
- 2 Show me the balance table. How do you know the groups started out alike?
- 3 Were errors clustered where you randomised, and does the interval clear our rule?

Question 2 tells you whether the comparison is fair. Question 3 tells you whether the result is enough to act on.

What today gives you

- Before-and-after** folds in everything else that changed · here it understated
- With-and-without** compares groups that already differed · here it overstated
- A lottery** makes groups alike on average, and more so the more units it draws
- Reading the result** a difference of two averages; judge its interval against the rule

Tomorrow: GreenWaste without the lottery. If the groups were not drawn, **was the gap between them at least stable before the programme started?**